



FIRST SEMESTER 2026-27
COURSE HANDOUT

Date: 02.08.2026

In addition to part I (General Handout for all courses appended to the Time table) this portion gives further specific details regarding the course.

Course No	: BIO F313
Course Title	: Animal Physiology
Instructor-in-Charge	: SHILPI GARG (shilpi@pilani.bits-pilani.ac.in)
Instructors	: Rajdeep Chowdhury (rajdeep.chowdhury@pilani.bits-pilani.ac.in) Ritobrata Bhattacharyya, Ananya Bhatt

1. Course Description:

Basic functional processes in animals; like homeostasis, membrane physiology, digestive, respiratory, excretory, circulatory, reproductive, nervous, sensory, muscular, endocrine systems will be studied at structural, functional, cellular and molecular level.

2. Scope & Objective:

This course attempts to bring the awareness among students for major features of animal physiology and co-relate that to the human physiology. Emphasis will be on structure and function relationship and importance of every organ. In addition, the course will also discuss about some of the diseases that are associated with complication in different organs.

3. Course learning outcomes:

- Integration of multiple organ systems to regulate and maintain homeostasis between the internal and external environments.
- Explore the process of enzymatic digestion and its physiological outcomes within the digestive tract.
- Functional mechanisms of the human kidneys in regulating fluid homeostasis and urine formation.
- Overview of the cardiac cycle, electrocardiogram (ECG) interpretation, types of circulatory vessels, regulation of systolic and diastolic blood pressure, and the various cellular components of blood.
- Anatomical and functional organization of nervous system, including its structural and spatial characteristics.
- The role of hormones in regulating homeostasis and supporting essential physiological functions.
- Understanding gas partial pressures, intrapleural pressure dynamics, lung volumes and capacities, respiratory disorders, and the oxygen-carrying capacity of hemoglobin.
- Study of sarcomere structure, the molecular mechanism of muscle contraction, the physiological basis of rigor mortis, and the phenomenon of muscle twitching.
- Exploration of the regulatory mechanisms governing gametogenesis, fertilization, and progression of pregnancy.

5. Text Books: Sherwood, L., Klandorf, H. and Yancey, P.H., Animal Physiology: From Genes to Organisms, 2nd edition, 2013, Brook/Cole Cengage Learning, Singapore

Reference Books: Sherwood, L. Principles of Human Physiology, 2007, Brook/Cole Cengage Learning,



5. Method of Conduct of the Course

The Animal Physiology course will be delivered through a combination of interactive lectures, tutorials (including lab based components during the tutorial hours), and group discussions to promote both conceptual understanding and hands-on experience.

Lectures: Core physiological concepts will be taught through structured classroom lectures, supported by visual aids such as diagrams, animations, and clinical case examples to enhance comprehension.

Tutorials and Group Discussions: These sessions will focus on clarifying doubts, exploring clinical correlations, and encouraging critical thinking through problem-solving activities and peer interactions. In addition, during some of the tutorial classes, practical sessions will be carried out to examine/ prepare cross-sections of fixed tissues and performing tissue staining to facilitate a detailed understanding of the cross-sectional anatomy of various organs, including the heart and kidney.

Assessments: The course will include periodic quizzes, written assignments, viva voce, and mid-term and end-term examinations to evaluate student progress and understanding.

Self-directed Learning: Students will be encouraged to engage in independent study through guided reading, online resources, and optional assignments to deepen their knowledge beyond the classroom.

6. Course Plan:

Lecture No.	Learning objective	Lecture session	Reference	Learning Outcome
1-2	Introduction to Physiology	Cellular Organization; Homeostasis: Basic Mechanisms	Ch# 1 TB	Coordination of different organ systems to maintain balance between external and internal environment.
3-7	Digestive system	Parts of the Digestive System and their function	Ch# 14 TB	Learn more about enzymatic digestion of food and their outcome in the digestive tract
8–11	Excretory system	Organs, Filtration, Re-absorption and Secretion	Ch# 12 TB	Operational process of human kidneys, fluid homeostasis and urine formation
12-17	Circulatory system	Circulatory Fluids, Heart, Cardiac Cycle, Regulatory Factors	Ch# 9 TB	Cardiac cycle, ECG, circulatory vessels, systolic and diastolic pressure, different cell types present in blood
18-25	Nervous System and Neuronal Physiology	Central and Peripheral Nervous System, Neurons	Ch# 4-5 TB	Structural and spatial characterization of nervous system and its function
26- 28	Circadian Rhythm & Endocrine System	Biological Clock, Hypothalamus, Pituitary, Glands and Hormones	Ch# 7 TB	How hormones perform body homeostasis and other vital functions



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

29-33	Respiratory system	Organs Involved, Mechanics, Gas Exchange and Transport	Ch# 11 TB	Partial pressure of gases, Intra-pleural pressure, lung capacity, respiratory disorders, hemoglobin and its oxygen carrying capacity
34-37	Muscular System	Skeletal, Cardiac and Smooth Muscle	Ch# 8 TB	Mechanism of muscle contraction, Rigor mortis, muscle Twitching
38-41	Reproductive System	Male and Female Reproductive System	Ch# 16 TB	Understanding gametogenesis, fertilization & pregnancy.

7. Evaluation Scheme:

Component	Duration	Weightage %	Date & Time	Remarks
Mid-Semester Test	90 Min.	25	10/10 FN1	CB
Assignment/in-class Quizzes/ Tutorial Activity and Assessment (Surprise/Announced)	-	15		CB/OB
Course Quizzes (2) (Announced)	-	20	TBA	CB
Comprehensive	3 Hrs.	40	15/12 FN	CB and OB

*Material in soft-copy will not be allowed in any evaluation component. For Open-Book Exam you are allowed to bring only the indicated books and your own handwritten notes. Printouts/xerox of lecture slides and class notes will NOT be allowed

8. Grading Policy:

Award of grades would be guided by the histogram of marks and course average. If a student obtains below 30% of the average of the top three performers or 40% of median, whichever is lower, the student may not be awarded a valid grade and the case will be reported as “NC”.

9. Chamber consultation hour: Will be announced in the class

10. Notices: Announcements regarding the course will be made in the lecture and will be displayed on Nalanda or will be communicated through email.

11. Academic Conduct Policy: It is expected that all students should attend the classes regularly and follow the highest standard of academic practice when participating in any evaluation component. Having a zero-tolerance for academic dishonesty, any case of misconduct, however minor, will be dealt with strictly with necessary actions and the case may be reported to the Dean, Instruction Division, if necessary.

12. Make-up policy: No make-up will be granted without a genuine case. A certificate from authenticated doctor must accompany the makeup application (only prescriptions/ vouchers for medicines will not be sufficient). Prior permission from IC is strongly recommended. No makeup for in class assignments and quizzes.

Instructor-in-charge
BIO F313